

WG16 - Space applications and fuels

Nuclear propulsion



Author	E-mail address	Personal Code
Curti Gianni		
De Sio Lorenzo		
Ferrara Federico		
Giussani Riccardo		
Guffanti Chiara Alessia		
Jance Sara		
Pitussi Francesco		
Piunti Matteo		
Pratola Oscar		
Segala Andrea Giulia		
Spinelli Giulia Leda		

Course of Introduction to Nuclear engineering
School of Industrial Engineering
Academic Year 2023-2024

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Acronyms

1 | Introduction

1.1 Brief overview over Nuclear Propulsion Systems

Nuclear energy in space propulsion systems has long been seen as a mean to enable missions in outer space that wouldn't be possible by any current chemical-based system. The main trouble related to space exploration is the need of a high-power output for long time capable of handling high mass flow rates. Nuclear propulsion systems provide high enough thrust to reduce the propulsion action and reach the maximum velocity sooner and high specific impulse to save propellant and allow higher peak velocity for the same propellant mass ratio. In the last century, nuclear rocket propulsion programs were initiated by the United States and Russia: these trials were successful to a large extent, however due to a variety of reasons nuclear-related programs were never further explored. Nevertheless, these engines proved to be twice as efficient as the most advanced chemical engines, allowing quicker transfer time.

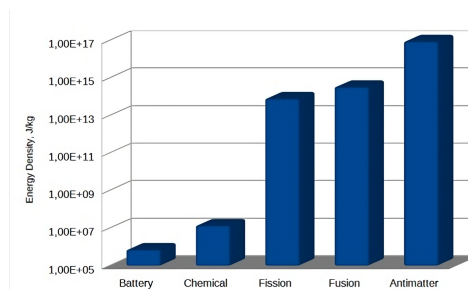


Figure 1.1: Energy density for different power sources

1.2 Rocket Engine Fundamentals

Rocket engines operate by expelling a high temperature gas through a nozzle to produce a thrust that accelerate the spacecraft in the direction opposite, as a result to the Isaac Newton's third law of motion ("For every action, there is an equal and opposite reaction"). Thrust is defined to be the force produced by the rocket engine as a result of the variation of the exhaust gas momentum. Thrust power can be computed with the following equation:

$$T = \dot{m}_f \cdot v_e$$

Where $\dot{m}_f$ is the mass flow rate and v_e is the exhaust speed out of the nozzle

With the assumptions:

- No gravity related forces
- No drag related forces
- Constant thrust
- Constant mass flow rate
- Constant specific impulse

From the conservation of the momentum, the total time rate of change momentum of the rocket plus the exhaust is equal to zero, applying Newton's second law of motion and rearranging the obtained equation, we can define a new term called specific impulse:

$$I_{sp} = \frac{T}{g_0 \cdot \dot{m}_f}$$

Where: T = total engine thrust g_0 = conversion factor $\dot{m}_f$ = mass flow rate The specific impulse is a very useful parameter in determining the efficiency of a rocket engine. Restarting from the previous equation and rewriting the

differential portion, we can obtain the expression for the maximum velocity increment achievable for a given vehicle mass fraction and engine-specific impulse:

$$\Delta v = g_0 \cdot I_{sp} \cdot \log(f_m)$$

Where f_m represents the vehicle mass fraction (m_0/m_f). This expression is known as the rocket equation or Tsiolkovsky equation. By combining the equations shown above (thrust power, specific impulse, Tsiolkovsky) it is possible to start making simple calculations to fully get the importance of Nuclear propulsion. Considering the two-body Keplerian Model it is possible to commute the escape velocity of the Earth through the following equation:

$$V_{esc} = \sqrt{\frac{2\mu}{R}} = 11.2 \frac{km}{s}$$

Where μ is defined as the standard gravitational parameter that emerges by applying the Keplerian Model. In our case we are interested in the earth escape velocity, so $\mu = 398600.4 \frac{km^3}{s^2}$ and $R = 6378 km$. As a consequence it is also possible to calculate the kinetic specific energy:

$$E_k = \frac{1}{2} \cdot V_{esc}^2 \approx 60.5 \frac{MJ}{kg}$$

And now by applying the Tsiolkovsky equation and the specific impulse we find out that we can obtain the equivalent amount of energy from either 6 kg of chemical source either 60 mg of nuclear fission related source. However the numbers do not correctly portrait reality, but it is worth noticing that, despite the oversimplified problem, the comparison between the masses obtained is a good starting point to further prove the effectiveness of nuclear space propulsion. This result depends on the different values of energy density for each given source. Down below there is a comparison amongst different fuels:

Fuel	Energy Density (J/kg)	Notes
Battery	$7.2 \cdot 10^5$	Lithium Ion
Chemical	$1.4 \cdot 10^7$	LO_2/LH_2
Fission	$8.2 \cdot 10^{13}$	U^{235}

Table 1.1: Energy density comparison[1]

2 | Distinction between NTP and NEP

2.1 Distinction between NEP and NET

2.1.1 NEP and NTP

NEP (Nuclear Electric Propulsion) and NTP (Nuclear Thermal Propulsion) are two different methods in which the nuclear power is employed in production or acceleration of propellant for the thruster of spaceships and space rockets. In NEP the electricity is produced by a nuclear reactor mounted on the rocket itself and is necessary in the first step for ionising an inert gas propellant by heating it, then for the generation of the electric field or magnetic field that accelerate that gas through a nozzle creating a low amount of thrust-pushing the spacecraft forward. The major propellants employed in NEP are Hydrogen, Lithium and Xenon for their low molecular weight. This technology has order of magnitude less thrust of NTP.

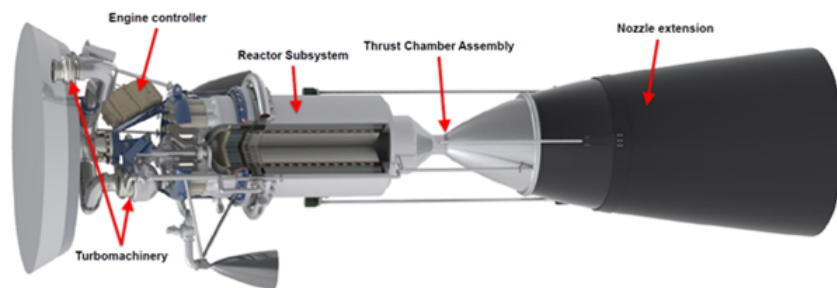


Figure 2.1: Scheme of a NTP engine

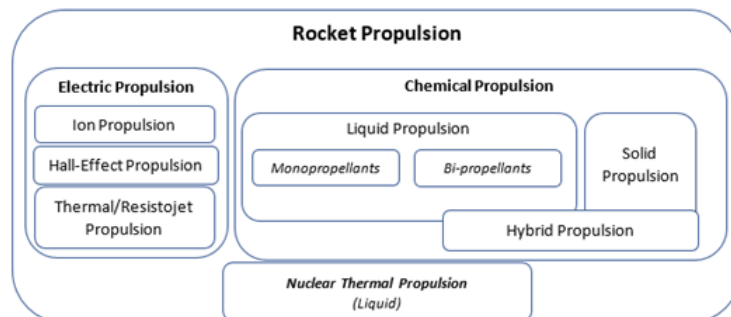


Figure 2.2: Examples of rockets' propulsion

The Nuclear Thermal Propulsion instead directly utilises the thermal energy generated by nuclear fission in a nuclear reactor. It uses a liquid propellant (hydrogen), which is pumped through the reactor core, where it rapidly heats up and expands to a gas and then is forced out of the rocket nozzle creating high thrust and accelerating the spacecraft.

2.1.2 Why NTP?

Today's best chemical propulsion systems can achieve ISPs of 465 seconds, while NTP can achieve almost two times the ISP of 900 seconds. In addition to the high ISP compared to other propulsion systems, NTP has an additional benefit of having a high thrust (10-15 klbf) to weight ratio so it dramatically reduces IMLEO (Initial Mass in Low Earth Orbit), the required number of SLS (Artemis' Space Launch System) launches and enables "affordable Mars Missions" not possible using other propulsion options. NTP will enable a much shorter round trip in Mars mission so there is a significant reduction in crew exposure to space radiation and the debilitating physical effects of prolonged exposure to a zero-g environment. The use of an NTP system allows vehicle reusability and component commonality; also allows artificial gravity operation and increased abort capability. If an NTP system is used individually or in a

clustered arrangement, it allows many other mission applications (e.g., reusable lunar cargo delivery, crewed lunar landing missions, crewed asteroid missions, a high energy injection stage for shortened robotic science missions to the outer planets).

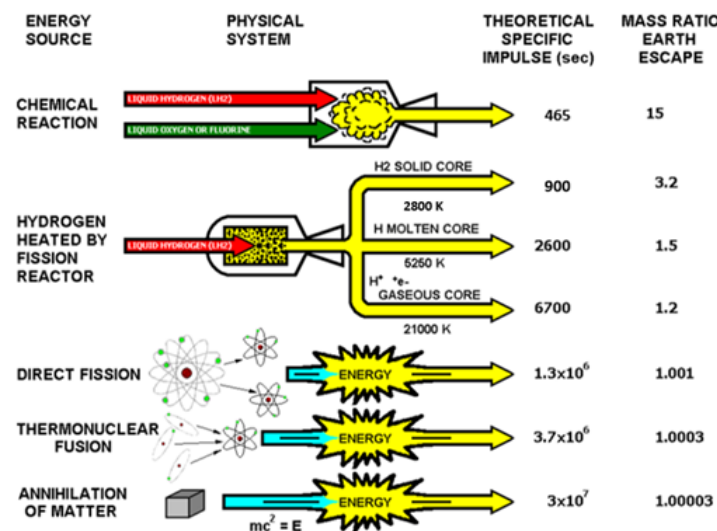


Figure 2.3: Comparison between different kind of energy sources for propulsion

2.1.3 HYBRID NEP

Recent studies have explored the possibility of using an NEP system combined with a chemical propulsion stage for human missions to Mars. The chemical stage would perform the high thrust required for climbing out of and dropping into planetary gravity wells with the NEP stage providing the balance to complete the mission. Results of analyses to date indicate that NEP-Chem hybrid vehicles can provide mission profiles comparable to Nuclear Thermal Propulsion (NTP) systems with smaller, lower-temperature reactors.

2.2 NERVA

Nerva (Nuclear Engine for Rocket Vehicle Applications) was an experiment made in the mid of twentieth century (after the second world war) by some engineers whose efforts were spend in take advantage of the massive power of atomic fission for aircraft and missile propulsion. After several test, in which were founded some issues as the required shielding for the crew or the fear of radiation at crash sites, in 1955 the Project Rover was born with the help of the Atomic Energy Commission (AEC). The main achievement of the project was the development of reactors for nuclear rockets. In 1959 the NASA entered in the Project and the new target was the development of a nuclear rocket for long-duration space flight. The first NERVA NRX test was run in September 1964 in Nevada. In 1969, the AEC successfully tested a second-generation NERVA (XE) engine dozens of times. Funding for NERVA, however, decreased in the late 1960s and the program was cancelled in 1973 before any flight tests of the engine took place. The biggest achievement of the NERVA project was the realization of the Hydrogen pumping system.

NERVA'S HYDROGEN PUMPING SYSTEM

A nuclear rocket engine achieves its thrust by heating hydrogen to temperatures in the 1650 to 2200 °C range and expanding this gas through a nozzle. Liquid hydrogen in the vehicle tank stored at -250°C is pumped to engine operating pressures by a turbopump. The hydrogen then passes through the tubes of a regeneratively cooled nozzle into the core, where it is heated to outlet temperatures. Passing through the nozzle, the hydrogen expands and accelerates to supply the engine thrust needed to push the rocket into space. The engine, which is 22 feet high from the top flange to the exhaust exit of the nozzle, has a performance goal to provide 50,000 pounds of thrust. The reactor, which is a right circular cylinder approximately 3 feet in diameter by 4 feet high enclosed within the pressure vessel, produces a thermal power of about 1000 mW. The annulus between the reactor core and the pressure vessel is occupied by a beryllium neutron reflector, which contains the reactor control drums. Directly above the inlet

end of the reactor is a radiation shield that shadows the engine 68 components, vehicle, and payload, and protects them from excessive radiation doses and heat deposition. Liquid hydrogen passes through the turbopump into the propellant piping and into the regenerative cooling tubes of the nozzle. The spherical tanks supply actuator gas for engine startup. The hydrogen extracts the heat generated in the reflector by neutron and gamma absorption and attenuation and cools the various parts of the reflector, so that the hydrogen is gaseous when it enters the shield region. After passing through the shield, the hydrogen enters the core. In the core, uranium fuel is dispersed in graphite elements that are pierced by circular propellant channels. The flow through the channels is controlled by orifices to obtain uniform temperature rise across the reactor. The heat generated by the fissioning of the uranium heats the hydrogen to an exit temperature significantly more than 1650 °C.

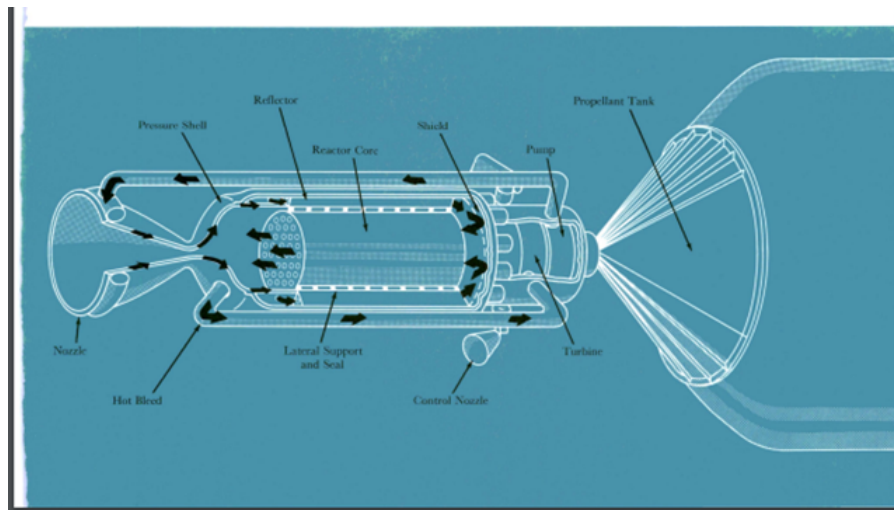


Figure 2.4: How works the NERVA's Hydrogen pumping system

3 | NTP architecture

The reactor is a subcomponent housed within the engine assembly which acts as a heat exchanger to transfer the energy from fission to heat the propellant to high exit temperatures ($T > 2350\text{ K}$), the propellant is subsequently expanded through a nozzle to provide thrust. The reactor design must incorporate the components necessary to enable controllable criticality under proposed operating conditions. The critical aspects of this design include the fuel and moderator design within the reactor active core to enable high operating temperatures and criticality under nominal conditions. Ultimately, performance of the reactor system is limited by the reactor exit temperature [2]. The main elements of a rocket nuclear reactor are similar to the ones of a ground-based reactor with some modification in order to fit the reduced space available on-board. The main elements are:

radiation shield material that stops radiation from leaking out to harm structure or crew.

reflector reduces leakage by scattering neutrons back into the core.

moderator low neutron absorber that reduces the probability of neutron leakage and also reduces the energy of fast neutrons to the level of slow, or thermal, neutrons so that they may support the fission reaction. Materials include hydrogen, carbon, beryllium, and heavy water.

fuel enriched uranium, containing U233, U235, or other fissionable fuel, such as plutonium P239.

propellant Hydrogen is the propellant of choice

control rods material with a high thermal neutron absorption cross section, which may be inserted into or withdrawn from the core to alter the progress of the fission reactions. Materials include boron, tantalum, and cadmium.

The criticality of the reactor is characterized by the factor k , which is the ratio of the number of neutrons present at the end of any particular neutron-generating event to the number of neutrons at the initiation of the generation event [3].

3.1 Solid core reactors

3.1.1 Fuel arrangement

The ROVER or NERVA type systems used solid core reactor technology with fuel elements in rods as depicted schematically in Figure 3.1. The hexagonal fuel rods were composed of a graphite substrate with uranium carbide and zirconium carbide dispersed within it. The graphite acted as the moderator for the nuclear fission reactions. A number of coolant channels were distributed in the fuel rod through which the propellant would pass, picking up heat from the nuclear reactions proceeding within the fuel rods. The hexagonal fuel rods, which are around a meter or

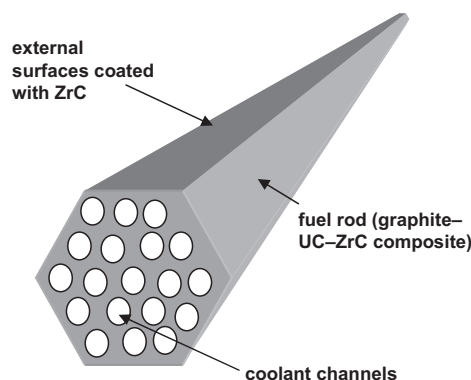


Figure 3.1: Fuel rod

two in length, are closely packed to form the solid core of the reactor and require the inclusion of support elements to support the reactor structure, as shown in Figure 3.2. The support elements require thermal and nuclear radiation protection, which led to the complexity of their design.

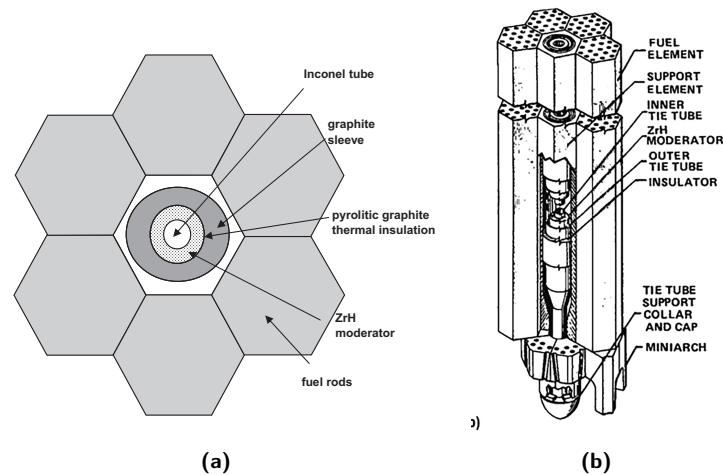


Figure 3.2: Packed fuel

3.1.2 Criticality control

Control of the criticality of the reactor in the solid core reactor for propulsion applications involves a set of rotatable control drums distributed in sectors around the reactor core, as shown in Figure 3.3. The sectors are composed of reflector material such as beryllium and each contain a drum that has a sector composed of a segment of a neutron-absorbing (or “poison”) material such as boron. To start the reactor, the so-called poison plate is rotated such that only reflector material faces the core. Thus neutrons are reflected back into the core to continue the nuclear reaction. As the temperature builds, the drums may be rotated so that the poison plate can capture neutrons instead of reflecting them back into the core. The positioning must be controlled so that the reactor remains critical but does not become unstable.

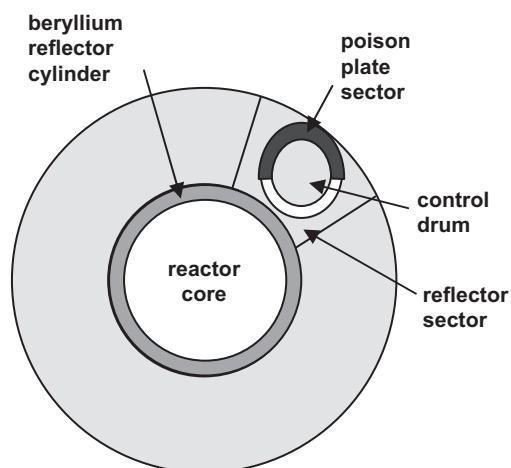


Figure 3.3: Control drums

3.1.3 Performances and criticalities

The thrust to weight ratios for such reactors fall in the range of 3 to 12. A typical solid core nuclear rocket can operate with a gas temperature of about 2800K at around 3.5 MPa, issuing through a nozzle of expansion ratio

equal to 40, thereby producing a specific impulse $I_{sp} = 800 \text{ s}$. This is almost twice the value for the most advanced chemical rocket motor.

Prototype reactors have been tested, and the design of a flight weight engine has been accomplished. Thus the solid core nuclear thermal propulsion system is a near-term option for interplanetary flight. Obviously, the strength of the structural components exposed to high temperatures for extended time periods is an issue of concern in the design process.

3.2 Particle bed reactors

A refinement of the solid core reactor is the particle bed reactor concept, which uses coated fuel particles in a packed bed through which the propellant flows, cooling the reactor and heating the propellant for acceleration through a nozzle. Evidences show that the major limitations of the proven solid core reactor in space applications is the lower thrust to weight ratio with respect to the latest chemical rockets. This is true because the solid core reactor is handicapped by the long narrow coolant passages in the fuel rods. In fact:

- The relatively low surface area to volume ratio of the passages limits the heat transfer to the propellant passing through them.
- The flow rates of propellant are relatively low because of the large pressure drop associated with the small-diameter passages.

Thus, the following design changes were introduced:

- To improve the surface to volume ratio and thereby improve the heat transfer characteristics of the reactor, the fuel element was formed by packing small-diameter (0.4 mm) particles into the annular space between two porous tubes. This provides a ratio of fuel surface area to particle bed volume of about $100 \text{ cm}^2/\text{cm}^3$. The fuel particles are spheres of UC_2 encapsulated by graphite and finally coated with zirconium carbide. Encapsulation serves to avoid the loss of fission products. The hydrogen gas propellant flows into the annulus of the fuel element, passing over the particles and out of the central core and to the nozzle.
- Instead of having the propellant flow axially down the channel in a fuel rod it flows radially through the packed bed of fuel spheres and into the central channel, which directs it to the nozzle. This permits less pressure drop and therefore higher propellant flow rates.

Both improvements allow construction of a reactor with a much higher power density. The fuel elements are encased in a hexagonal block of moderator material, as shown in Figure 3.4.

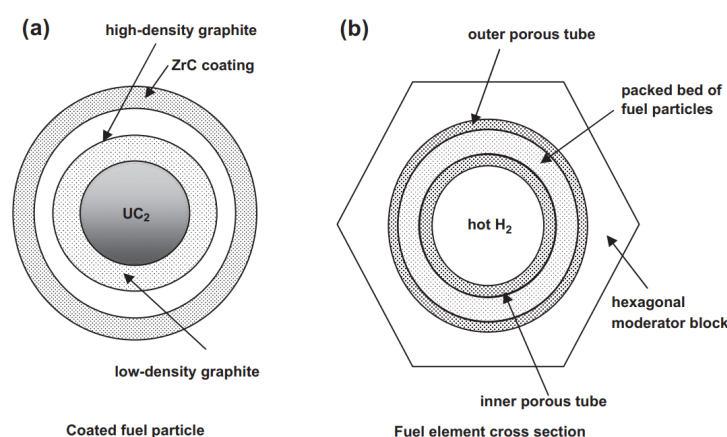


Figure 3.4

A number of these structures are closely packed within the rocket motor casing, as shown in Figure 3.5, and the hot gas flowing out of each is directed to the nozzle.

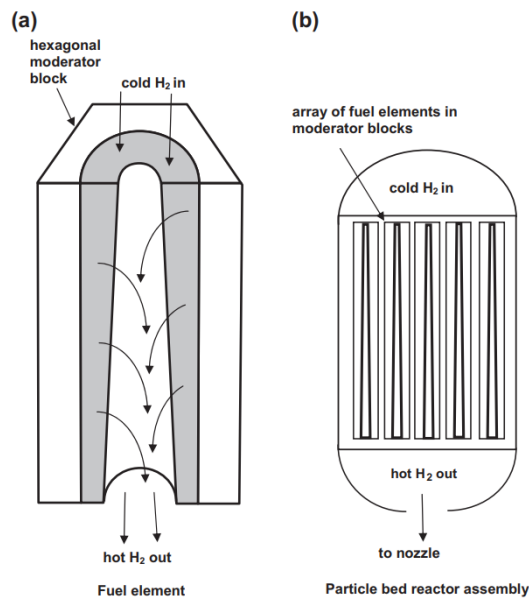


Figure 3.5

A third improvement is the use of lithium hydride as the moderator material. The LiH has a low density (approximately 800 kg/m^3) and can withstand relatively high temperatures (around 1000 K). The particle bed reactor theoretically makes possible a nuclear rocket with $T/W = 40$ and $I_{sp} = 1000 \text{ s}$.

3.3 Feed system

Candidate propellant feed system cycles typically include the following: a gas generator system, which uses combustor gas generator to drive the feed turbopump; a staged combustion system, which uses combustion products from a separate preburner to drive the feed turbopump; and an expander cycle, which uses the reactor heated propellant to drive the feed turbopump directly. The expander cycle shown in Figure 3.6 is considered the best alternative for a solid core nuclear thermal rocket motor.

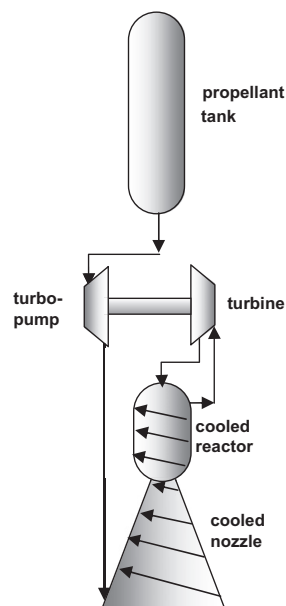


Figure 3.6: Feed system

4 | Hydrogen

4.1 Propulsion Advantages

Hydrogen is the propellant of choice for NTP, due to it having the lowest molar mass, which makes it the best propellant for a given stagnation temperature. By feeding it through the reactor core, the hydrogen heats up (for the propulsion purposes) and at the same time acts as a coolant for the reactor.

The advantages are clear when looking at the formula defining the specific impulse, one of the most important parameters in propulsion: it can be defined as the exit velocity of the propellant v_e , divided by the gravity acceleration on Earth g_0 . The exit velocity can be defined as:

$$v_e = \sqrt{\frac{2\gamma}{\gamma - 1} \frac{R}{M_{mol}} T_c \left(1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right)}$$

It is clear that increasing the temperature of the propellant and decreasing its molar mass brings an higher exit velocity and therefore and higher specific impulse.

Hydrogen also shows the useful property of being a good moderator: it is able to absorb slow neutrons and slow down the higher energy ones. On the ground it is not typically used due to it being gaseous at all but cryogenic temperatures, making hydrogen compounds such as water a more suitable choice.

4.2 Technical Drawbacks

While being very good for the strictly propulsive part, hydrogen is the cause of major problems to the overall system: High temperatures are desired to increase the specific impulse: while temperatures higher than 3000 K cause an higher probability of dissociation of hydrogen, reducing again the molar mass as desired, high temperature hydrogen causes erosion of materials such as graphite, requiring either a coating or a lower temperature. Existing NTP systems employ lower temperatures with respect to chemical propulsion systems, however the gain from the low molar mass of hydrogen still makes them better in terms of specific impulse.

Also, due to its cryogenic nature, hydrogen is hard to store, requiring extremely low temperatures and large volumes due to its low density. This makes it hard to store the rocket inside typical stages, and would require the development of new boosters and launch strategies.

Finally, long term storage is fundamental for the type of missions NTP is suitable for, making the reduction of boil-off one of the main focuses from the storage point of view: the fact that the tank is close to the reactor makes it necessary to have a shield to prevent the radiation leaking from the core from reaching the hydrogen tank.

5 | Challenges in operating NTP

Due to its peculiar chemical characteristics the use of hydrogen in the NTP engines can lead to several issues as rod corrosion and crack propagation in the nozzle. During the NERVA program was observed how the hydrogen flowing in the graphite control rods could penetrate the cracks reacting with the graphite. The occurring chemical reaction producing hydrocarbons causes the progressive corrosion of the rods and may also vary the performances, due to the change in the propellant composition and properties.

5.1 Rod Corrosion

5.1.1 Brief description of the phenomenon

The NERVA engine used a multilayered coating of the type (U,Zr)C/Gr positioned through chemical vapor deposition technique (CVD), but this led to crack formation once the coating had cooled down. The hydrogen flowing the channels would then penetrate the crack, reacting. The corroded would then leave the reaction site as hydrocarbon. From the literature [4], a viable solution may be a further coating of the previously described multilayered coatings, with a diffusion barrier (U,Zr)C-Gr/NbC (or ZrC) and a further overlayer made of Mo/Nb/ZrC/Mo. The concept is the following: the niobium carbide layer minimize the diffusion of carbon from the graphite substrate and fill its pores. Moreover the NbC and Mo coatings act as strain compliant coatings since the thermal expansion coefficient of Nb-Mo alloys lays between the one of the Gr and NbC/ZrC. In addition the diffusion of hydrogen is slower in molybdenum compared with what can be observed in the other materials and thanks to its ductility, during the molybdenum deposition all the cracks in the underlying layers can be sealed increasing the effectiveness of the diffusion barrier.

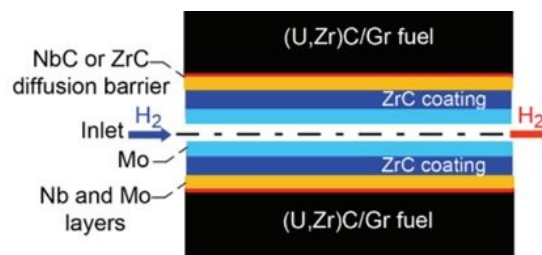


Figure 5.1: Multilayered coating in the rod channel

5.1.2 Conclusions

Experimental evidence have shown that it is possible to properly deposit the coating structure described above. In some cases, it has been observed how the Mo layers had partially debonded from their substrate. The origin of the phenomenon can be a high oxygen density in the Mo flow used in the deposition, so further development are require in order to avoid any oxygen contamination during the deposition process.

5.2 Crack propagation in the nozzle

The material of the nozzle should be accurately chosen since two main hydrogen related issues may occur. On one side a graphite nozzle may react with the hot hydrogen producing hydrocarbons and corroding in the process; on the other side metal nozzle may be affected by crack propagation due to the embrittlement phenomenon. Embrittlement is a complex phenomenon of which the mechanism is still under study, that is related to metal exposure to environment with a high concentration of hydrogen. One of the possible explanations is that the absorbed hydrogen recombines in molecules once in the voids of the material causing a pressure from the inside of it. Thus, the crack propagation is aided by the internal pressure. Further studies are required to find a proper coating which could both prevent the embrittlement and withstand the nozzle operative conditions

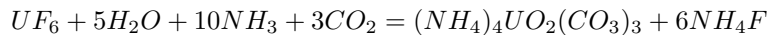
6 | Fuel production process

6.1 Uranium dioxide

There are four methods for converting UF_6 to ceramic grade UO_2 powder[5]:

6.1.1 Precipitation processes

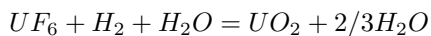
- Ammonium diuranate process (ADU): This process involves dissolving UF_6 in water and reacting it with ammonia to form a precipitate. The precipitate is then separated, dried, and reduced in a furnace to form UO_2 powder. After the furnace process, the powder may have to be slightly oxidized to make it stable in air. Milling, sieving, or both may also have to be performed to control agglomerate size. To improve pressability, the powder must be compacted and granulated by crushing through a coarse sieve. A small amount of lubricant, usually zinc stearate, is then added to decrease internal friction during the pressing of the pellets.
- Ammonium uranyl carbonate process (AUC): This is used to reduce the number of steps in the ADU process and to obtain a precipitate with a lower fluorine content. Gaseous UF_6 is reacted with a solution containing ammonium carbonate:



that produces crystalline AUC as a precipitate. Because of its high solubility an additional precipitation step is performed before the solution goes to the final effluent treatment. After filtering, the AUC is treated in a fluidized-bed-type furnace in which drying, reduction of fluorine content, and reduction to UO_2 with hydrogen are performed. After a slight oxidation in a separate step, the free-flowing powder can be used directly for pellet pressing.

6.1.2 Dry processes

Gaseous UF_6 is brought into an atmosphere of super heated steam and reacted to solid UO_2F_2 . The uranylfluoride is then reduced by hydrogen to UO_2 , which is the stable oxide in this reaction. The overall reaction can be written as:



Two different routes of the dry process have been developed:

- Fluidised bed route dry process: The fluidized bed route process for converting UF_6 to UO_2 powder is a continuous process that takes place in a number of fluidized beds connected in series. In the first bed, the UF_6 is hydrolyzed with steam to form UO_2F_2 . The UO_2F_2 is then reduced to UO_2 in the subsequent beds with hydrogen. The process temperature is typically around 600°C . The fluidized bed process has several advantages over other processes for converting UF_6 to UO_2 powder. First, it is a continuous process, which means that it can operate 24/7 and produce large quantities of UO_2 powder. Second, the fluidized bed process produces UO_2 powder with a high purity and a consistent particle size. Third, the fluidized bed process is relatively simple to operate and maintain. However, the fluidized bed process also has some disadvantages. First, it is a more complex process than other processes, such as the rotary kiln process. Second, the fluidized bed process requires more energy than other processes. Third, the fluidized bed process produces more dust than other processes.
- Rotating kilns dry process: This is a batch process that takes place in a rotating kiln. The UF_6 is fed into the kiln at one end and the UO_2 powder is collected at the other end. The kiln is heated to a temperature of around 600°C . The hot gases from the kiln are used to dry the UO_2 powder and to remove any remaining fluorine. It's a simpler process than the fluidized bed process. However, it also has several disadvantages. First, it is a batch process, which means that it cannot operate 24/7 and produces smaller quantities of UO_2 powder. Second, the rotary kiln process produces UO_2 powder with a lower purity and a less consistent particle size than the fluidized bed process. Third, the rotary kiln process requires more time to complete than the fluidized bed process.

6.2 Uranium carbides

Uranium carbide fuels have several important features [6], including high thermal conductivity, high operating temperatures, and stability in hydrogen. They also have low melting points, which can be raised by adding other carbides. However, they are non-metallic with low ductility, which can lead to cracking under severe thermal cycles. Processing and shaping these materials can also be problematic due to their high melting points and hardness. There are two main steps to sintetize Uranium carbides:

6.2.1 Carburization of UO_2 powder

There are a variety of methods for carburizing UO_2 powder to form uranium carbide. Some of the most common methods include:

- Solid-state carburization: This process involves heating a mixture of UO_2 powder and carbon powder to a high temperature (typically around 1800°C) in a vacuum or inert atmosphere. The carbon atoms diffuse into the UO_2 lattice, forming uranium carbide.
- Liquid-phase carburization: This process involves heating a mixture of UO_2 powder and carbon powder in a molten metal bath, such as molten uranium or molten iron. The carbon atoms diffuse into the UO_2 lattice, forming uranium carbide.
- Plasma carburization: This process involves exposing UO_2 powder to a plasma discharge. The high energy of the plasma causes the carbon atoms to dissociate and diffuse into the UO_2 lattice, forming uranium carbide. The choice of carburization method will depend on the desired properties of the final uranium carbide product. For example, solid-state carburization is typically used to produce high-purity uranium carbide, while liquid-phase carburization is typically used to produce uranium carbide with a specific grain size or microstructure.

6.2.2 Consolidation of uranium carbide powder

There are also a variety of methods for consolidating uranium carbide powder into the desired shape and size. Some of the most common methods are:

- Hot pressing: This process involves compressing uranium carbide powder in a die under high pressure while applying heat. The heat causes the uranium carbide particles to bond together, forming a dense solid.
- Cold pressing and sintering: This process involves compacting uranium carbide powder in a die under high pressure and then sintering the compact at a high temperature. The sintering process causes the uranium carbide particles to bond together, forming a dense solid.
- Liquid-phase sintering: This process involves heating a mixture of uranium carbide powder and a liquid phase material to a high temperature. The liquid phase material melts and wets the uranium carbide particles, allowing them to bond together more easily.

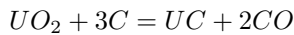
The choice of consolidation method will depend on the desired properties of the final uranium carbide product. For example, hot pressing is typically used to produce uranium carbide components with high density and strength, while cold pressing and sintering is typically used to produce uranium carbide components with complex shapes.

6.3 Uranium nitrides

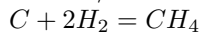
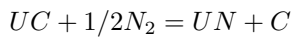
Uranium nitride fuels (UNF) [7] are proposed for space power reactors and liquid-metal-fast-breeder reactors due to their superior performance. UNF has the most favorable properties and potentially the best performance characteristics of the conventional ceramic fuels.

6.3.1 UN powder sinteting process

The UN synthesis process is a sequential conversion process that combines the carbothermic-reduction process used to fabricate carbide fuels with a modification of the UO_2 -to-UN conversion process. The first step is to heat UO_2 plus carbon powders to form UC according to the following general reaction:



After this reaction is partially complete, $N_2 - 6H_2$ gas is introduced into the furnace to create UN and remove the free carbon:



This process produces high-quality UN powder that is suitable for fabrication into reactor-grade fuel pellets. It is also relatively inexpensive, as it does not require the use of high-purity uranium metal.

6.3.2 UN pellet fabrication process

The UN pellet fabrication process involves the following steps:

- Powder preparation The as-synthesized UN powders are vibrationally ball-milled to reduce particle size and increase sinterability. Ball milling is a high-energy process that uses hardened steel balls to grind the powder particles into smaller pieces. This increases the surface area of the particles, which improves their ability to bond together during sintering. The milling time must be carefully controlled, as too much milling can make the powder too fine and difficult to press into pellets. On the other hand, too little milling will result in pellets with low density and poor physical properties.
- Pellet pressing Once the powder has been milled to the desired size, it is slugged, granulated, and pressed into green pellets. Slugging is a process of compacting the powder into a solid cylinder, or slug. Granulation involves breaking the slug into smaller pieces, or granules, which are then pressed into pellets. The pressing pressure must be high enough to consolidate the powder particles into a solid pellet, but not so high that the pellet is damaged. The pellets are typically pressed in a die set using a hydraulic press.
- Sintering The green pellets are sintered to achieve the desired density and grain size. Sintering is a process of heating the pellets to a high temperature, but below the melting point. At this temperature, the powder particles bond together to form a dense, solid structure. The sintering temperature and time must be carefully controlled to avoid over-sintering, which can cause the pellets to shrink and crack. Over-sintering can also lead to the formation of large grains, which can weaken the pellets.

6.3.3 Fuel pins fabrication process

The fuel pins are fabricated using Nb-1 Zr cladding and a chemical-vapor-deposition (CVD) tungsten liner to separate the fuel and the cladding. The fuel stack contains 12 pellets with a depleted uranium UN insulator pellet on each end of the stack. The steps to produce the pins are:

1. Machine and clean the components.
2. Vacuum anneal the Nb-1 Zr at 1775 K for 1 h.
3. Electron-beam(EB)-weld the bottom end cap to the cladding.
4. Load the tungsten liner, fuel pellets, and spring.
5. EB-weld the top end cap.
6. Evacuate and backfill the pin with high-purity helium.
7. Seal the pin with a laser closure weld.
8. Vacuum-anneal the finished pin at 1475 K for 2 h.
9. Inspect.
10. Ship the pins to HEDL for assembly into the SP-1 test vehicle.

7 | Environmental concerns

7.1 Environmental safeguards

The risks associated with employing nuclear power in space are similar to those encountered on Earth. Radiation safety is provided in two ways:

- 1) The basic approach to safety in orbit relies on moving the spacecraft into a stable, long-term storage orbit, close to circular, at a height of more than 530 miles. There, nuclear reactor fission products can decay safely to the level of natural radioactivity or they can be transported away from Earth sometime in the future.
- 2) The back-up emergency approach involves the dispersion of fuel, fission products and other materials with induced activity into the upper layers of the Earth's atmosphere. During the descent, aerodynamic heating, thermal destruction, melting, evaporation, oxidation, etc., are expected to disperse the fuel into particles that are sufficiently small as to pose no excess radiological hazard to Earth's populations or to the environment.

The backup safety system was introduced after the failure of the change in orbit of the Cosmos-954 spacecraft. The descent of the Soviet Union's spacecraft resulted in large radioactive fragments of wreckage being strewn across a thin strip of northern Canada in 1978.

A few accidents occurred during the 1960s and 1970s.

As a result, a United Nations working group has developed aerospace nuclear safety design requirements where:

- 1) The reactor shall be designed to remain subcritical if immersed in water or other fluids, such as liquid propellants;
 - 2) The reactor shall have a significantly effective negative power coefficient of reactivity;
 - 3) The reactor shall be designed so that no credible launch pad accident, ascent, abort, or re-entry from space resulting in Earth impact could result in a critical or supercritical geometry;
 - 4) The reactor shall not be operated (except for zero power testing that yields negligible radioactivity at the time of launch) until a stable orbit or flight path is achieved and it must have a re-boost capability from low-Earth orbit if it is operated in that orbit;
 - 5) Two independent systems shall be provided to reduce reactivity to a subcritical state and these shall not be subject to a common failure mode;
 - 6) The reactor shall be designed to ensure that sufficiently independent shutdown heat removal paths are available to provide decay heat removal;
 - 7) The unirradiated fuel shall pose no significant environmental hazard;
- and
- 8) The reactor shall remain subcritical under the environmental conditions of the postulated launch vehicle explosions or range of safety destruct actions.

7.2 Disposal Requirements

There are a number of disposal options for space nuclear reactors and associated risks, mostly in the long term, based on probabilities of Earth reentry. The results are based on a five year study that was conducted between 1978 and 1983 on the space disposal of high level nuclear waste, at Lewis Research Center. The study provides assessment of disposal options, stability of disposal or storage orbits, and assessment of the long term risks of the stuff coming back to Earth.

The disposal options available that might work best depend very much on how the nuclear reactor is going to be used in the mission scenario.

A spent reactor is a device that has been operated and is radiologically active at the end of life. Normally the end of life would occur after normal operations and the number of reuses that it has been designed for. But, the end of life could also occur from a disabling accident, in which case a disposal option may be required too.

Then the question is what to do with that spent reactor to eliminate or minimize the subsequent hazard of the radioactive material coming back to Earth: being released in the biosphere.

Here's a list of ten factors of consideration. If there is a need for disposal at the end-of-life, it's a fairly complex problem and needs to be considered in terms of trades:

- 1) Disposal destination;
- 2) Propulsion system requirements and cost;
- 3) Operational complexity, reliability and cost;
- 4) Failure modes and accident environments;
- 5) Payload response to environment;
- 6) Orbit evolution consequences of failed orbits;
- 7) Payload monitoring;
- 8) Retrieval/rescue mission capability;
- 9) Reentry probability - short vs long term;
- 10) Radioactive release risk to biosphere.

Let's talk about reentry risk. For example, a propulsion system failure might occur during injection prior to actually escaping the Earth.

Some disposal options include the stable solar orbit, the heliocentric planet-crossing orbit and then the moderate altitude Earth orbit. In a long term an Earth-crossing orbit looks like a fairly acceptable disposal place.

8 | Future perspectives

8.1 ESA

The ESA Future Space Transportation Systems department (STS-F) initiated two feasibility studies this year. These studies will identify key technologies to mature and intermediate steps to take to develop nuclear propulsion in Europe. Of course, the sine qua non conditions to use nuclear energy for propulsion is the robust implementation of the safety requirements from the early phases of the design. Following recommendations established in 2009 from the International Atomic Energy Agency (IAEA) and United Nations, the safety aspect is at the centre of the development of nuclear technologies for space applications. Project RocketRoll, or “preliminary european reCKon on nuclear electric pROpulsion for space appLications”, will explore the advantages of using a Nuclear Electric Propulsion (NEP) tug over classical propulsion systems for the demanding missions expected for the future of space logistics and exploration. It will identify main safety features for the design and highlight contingency and mitigation measures. Moreover, the study will determine what are the existing and missing key elements (technologies, modelling capabilities, testing facilities, etc.) in Europe that shall be matured for a nuclear electric tug operational after 2035. Project ALUMNI, or “preliminary elements on nuclear thermal propulsion for space applications” (ALUMNI), will explore the advantages of using a Nuclear Thermal Propulsion (NTP) over classical propulsion systems and highlight its potential profitability. More importantly, it will assess the key elements (technologies, modelling capabilities, testing facilities, etc.) and safety design features to make NTP possible.))) Nowadays, ESA (European Space Agency) is currently engaged in the Aurora space exploration programme. Aurora is ESA’s first step in human space exploration outside the low Earth orbits used for the Russian space station MIR and the International space station ISS. Through this programme ESA is elaborating a long-term plan for human exploration of space, with Mars as its main objective and the Moon a very likely intermediate step. The approved programme consists of two main elements: the “core programme” and the “robotic missions”. Among the tasks of the “core programme” is also the development of general enabling technologies for exploration. The first proposed mission of the “robotic missions” is the ExoMars mission, the first exobiology mission to planet Mars. Based on current physics knowledge, preliminary assessments and past experience, Europe will need to use radioisotope power sources for its missions to the outer Solar System. Europe might also need radioisotope heater units and possibly radioisotope thermoelectric generators for longer lasting lunar and martian surface missions, among which the ExoMars mission would be the first candidate.

8.2 TRL

Technology Readiness Levels (TRL) are different points on a scale used to measure the progress or maturity level of a technology. The scale ranges from 1 to 9, where TRL 1 is the lowest and TRL 9 is the highest. When a technology is at TRL 1, scientific research is beginning and those results are being translated into future research and development, while at TRL 9 the technology has already been proven to work during a flight mission in space. Each technology project is evaluated against the parameters for each technology level and is then assigned a TRL rating based on the project’s progress.

8.3 DRACO

NASA and the Defense Advanced Research Projects Agency (DARPA) will partner on the Demonstration Rocket for Agile Cislunar Operations, or DRACO, program. “With the help of this new technology, astronauts could journey to and from deep space faster than ever – a major capability to prepare for crewed missions to Mars” said NASA Administrator Bill Nelson. “The space domain is critical to modern commerce, scientific discovery, and national security. The ability to accomplish leap-ahead advances in space technology through the DRACO nuclear thermal rocket program will be essential for more efficiently and quickly transporting material to the Moon and eventually, people to Mars.” The goal of DRACO program is to demonstrate a nuclear thermal rocket (NTR) in orbit. NTRs use a nuclear reactor to heat propellant to extreme temperatures before exhausting the hot propellant through a nozzle to produce thrust. Compared to conventional space propulsion technologies, NTRs offer a high thrust-to-weight ratio around 10,000 times greater than electric propulsion and two-to-five times greater specific impulse (i.e. propellant efficiency) than in-space chemical propulsion. Phase 1 of the DRACO program involved two tracks. Track A conducted a baseline design of an NTR reactor. Track B developed an operational system concept to meet

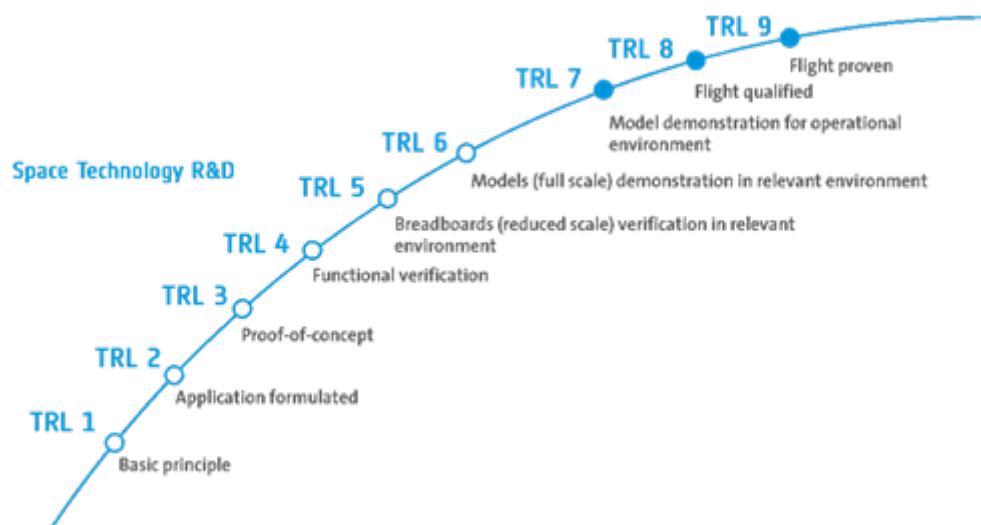


Figure 8.1: TRL

operational mission objectives and a demonstration system design that is traceable to the operational system but focuses on demonstrating the propulsion subsystem. Phase 2 and Phase 3 of the DRACO program will carry a single performer forward to the flight demonstration, which is envisioned to take place by FY27. This performer will be responsible for building the NTR and its demonstration system stage. Phase 2 will involve a cold flow test of the rocket engine without nuclear fuel. Phase 3 will involve assembly of the fueled NTR with the stage, environmental testing, and launch into space to conduct experiments on the NTR and its reactor.

8.4 Gas core reactors

Time in transit is a major issue for interplanetary space missions. In manned exploration missions, the duration of exposure to space radiation and microgravity conditions constitutes a health hazard for the crew (Sforza et al., 1993). Unmanned missions to asteroids, for example, to mitigate the hazard due to Earth-crossing asteroids, require fast response to ensure success (Sforza and Remo, 1996). To minimize flight time, high speed and therefore high thrust are required. At the same time, efficiency considerations demand low fuel consumption. The ultimate performance of a solid core nuclear reactor is limited by the loss of structural strength as temperature is increased. A gas core reactor uses the nuclear fuel in the gas phase, e.g., uranium hexafluoride. Although there is still the need for a moderating and reflecting encapsulation of the fissionable gas, heat transfer takes place between the hot fuel and the cool propellant while both are in the gaseous state. Typically some sort of inertial confinement of the gas is provided, usually in the form of a vortex flow. The gas core rocket is an open cycle system and therefore radioactive fuel is exhausted into the ambient. The application is for interplanetary missions in the deep space environment in which such emissions are acceptable. A typical gas core nuclear rocket can operate with a gas temperature of about 8600K at around 3.5 MPa issuing through a nozzle of expansion ratio $3 \frac{1}{4} 40$, thereby producing a specific impulse $I_{sp} \frac{1}{4} 2200$ s. Part of the increase can be attributed to the hydrogen being completely dissociated so that the effective molecular weight of the nozzle exhaust gas is approximately unity. This is almost five times the value for the most advanced chemical rocket motor. Attention here is focused on the open cycle reactor, a system that is limited in practical realization by one fundamental problem: effective containment of the uranium plasma. Containment must be one that minimizes the loss of the uranium fuel while reducing the thermal impact of the fission fuel on the structural components of the reactor chamber. Solution: Base bleed fuel confinement.

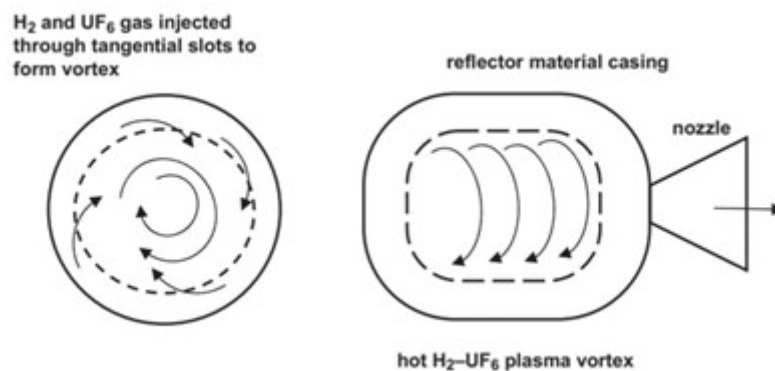


Figure 8.2: Gas core reactors

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